

Wool fiber—reinforced polymer composites

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2.1 Introduction

Wool-based composite materials are a type of composite material that utilizes wool fibers as the reinforcing phase within a matrix material (Santulli et al., 2013). These types of composites have several unique properties that make them attractive for use in various applications, such as automotive, aerospace, construction, and consumer products. The application of wool fibers as reinforcement in composite materials dates back to the early 20th century, although it was not until the 1960s that the use of wool-based composites began to increase significantly (Keya et al., 2019). This was due in part to the development of new methods for processing and blending wool fibers with various matrix materials, as well as the increasing demand for lightweight and high-strength materials in various industries. One of the primary benefits of wool-based composites is their high strength-to-weight ratio. Wool fibers are strong and flexible, and when combined with a matrix material, they can create a composite material that is lightweight yet strong and durable (Keya et al., 2019). This makes wool-based composites ideal for use in automobile and aerospace applications where weight is the critical factor.

In addition to their high strength-to-weight ratio, wool-based composites also have a number of other attractive properties. They are resistant to moisture and have good thermal insulation properties, which make them well suited for use in construction and other outdoor applications (Keya et al., 2019). Wool-based composites also have good acoustical properties and can be used to reduce noise transmission in a variety of applications. There are several methods for producing wool-based composite materials, including hand lay-up, filament winding, and injection molding (Alyousef et al., 2020). The choice of production method depends on the specific requirements of the application and the desired properties of the final composite material. One of the most common methods for producing wool-based composites is hand lay-up, which involves layering sheets of wool fiber and matrix material and then bonding them together using heat and pressure. Filament winding involves winding continuous fibers of wool around a mandrel to create a composite tube or pipe. Injection molding is a process in which a mixture of wool fibers and

matrix material is injected into a mold to create a shaped part. There are a number of different matrix materials that can be used in conjunction with wool fibers to create wool-based composite materials (Alyousef et al., 2020). These include thermosetting resins, such as epoxy and polyester, as well as thermoplastic resins, such as polypropylene (PP) and polyethylene.

Hence, wool-based composite materials are a type of composite material that utilizes wool fibers as the reinforcing phase within a matrix material. These materials have a number of attractive properties, including a high strength-to-weight ratio, good thermal insulation properties, and good acoustical properties. Wool composite can be produced using a variety of methods, including hand lay-up, filament winding, and injection molding, and can be made using a variety of matrix materials.

Wool is the fiber extracted from a sheep's or lamb's fleece, the hair of an Angora or Cashmere goat, or specialty fibers from a camel, alpaca, llama, or vicuna that has never been woven or felted. Wool is mostly sold as a by-product of sheep farms in the European Union, so it is wasted, burned, or dumped. More than 40% of contaminants are found in raw wool. A ton of raw material contains 640 kg of clean wool fiber, 150 kg of lanolin and dirt (soil), 40 kg of suint (wool grease, fats, and potassium), 20 kg of vegetable matter, and pesticide residues. Merino wool from the Oceania continent, which is the composition of Australia and New Zealand, predominates in European wool textiles. Australia accounts for 80% of fine Merino wool used in textiles, according to the International Wool Textile Organization. As a result, novel approaches to recycling coarse wool are required (Petek & Marinšek Logar, 2021).

Sheep wool is a renewable energy and a recyclable material with a low carbon footprint. Sustainable growth is also dependent on the valorization of local building materials, such as greasy wool, to reduce massive amounts of waste while preserving the location where it is produced, such as rural areas and their landscape. Utilizing wool waste as a natural, renewable, and biodegradable building material has a significant environmental benefit in terms of lowering CO₂ emissions and pollution (Parlato et al., 2022). Wool-based composite could be the best solution to this problem. It has various applications where it has some good characteristics compared to some materials. Various applications of wool-based composite could reduce the wastage of wool, thus reducing the usage of nonrenewable resources.

2.2 Fabrication of wool-based composites

2.2.1 Fabrication of polypropylene composites

The goal of the study by Kim et al. (2015) is to better understand how wool fibers, polymer melt flow index, and ammonium polyphosphate (APP) affect the performance of PP–wool composites as fire retardants. Cone calorimeter burn tests have been used to assess the flammability of composite materials with various formulas. Thermogravimetric analysis (TGA) was used in the study to look at the composites' weight reduction trend as the temperature increased. Tensile experiments have been

used to assess the impact of APP content on the mechanical characteristics of PP–wool composites. TGA was performed in the study to examine the weight loss pattern of the composites as the temperature rose. The effect of APP content on the mechanical properties of PP–wool composites has been evaluated using tensile tests. The experiment's findings showed that APP has a beneficial effect on mechanical characteristics because PP–wool–APP composites had higher tensile moduli (Kim et al., 2015). The reduced compatibility of wool fibers and PP resulted in lower tensile strengths for the PP–wool–APP composites than for PP–wool composites, but the reinforcing action of the wool in the composites still led to greater tensile strengths than PP–APP samples.

2.2.2 Fabrication of wool on flammability properties

Kim and Bhattacharyya (2016) investigated the impact of chemicals such as APP, talc, and maleic anhydride grafted PP (MAPP) types on the flammability and mechanical properties of extruded wool–PP composites on wool with additives. The study first compares the performance of wool–PP–APP composites as fire retardants using different APP types. A fire residue study performed for research purpose using an electron microscope resulted in exposing the morphology and chemical components of the char produced by the cone calorimeter test. The effects of APP, MAPP, and talc on the mechanical properties of wool–PP composites were also investigated. In this study, a fire dynamic simulator model was created to simulate the burning of the wool–PP composite in a cone calorimeter test. As a result, cone calorimeter studies revealed that MAPP-treated wool PP–APP composite has marginally higher peak heat release, total heat release, and smoke production rate than untreated wool–PP–APP composite (Kim & Bhattacharyya, 2016).

The primary goal of this inquiry is to individually investigate, using TGA and cone calorimeter, the thermal and individual flammability characteristics of waste-based biochar and wool (Das et al., 2017). Additionally, the individual mechanical characteristics of the biochar were assessed using the nanoindentation technique, whilst those of the wool were assessed using an Instron testing device. Furthermore, a cone calorimeter, limiting oxygen index (LOI), TGA, and Instron universal testing machine (UTM) were used in this study to assess the combustion, heat, and physical properties of the composites. Finally, using a scanning electron microscope (SEM), the microstructures of the surfaces that had been tensile cracked and the char that had resulted from the burning of composites were examined. According to Das et al. (2017), wool ignited quickly, but because of its advantageous char-forming capacity, the heat release rate (HRR) rapidly decreased once the peak HRR was attained.

The goals were to investigate the stimulatory activity of wool and natural fiber hybridization on intumescent PP/kenaf fiber (KF) composites based on APP and to study the performance of the resultant fusion (PP/KF/APP/wool fabric [WF]) composites (Subasinghe et al., 2018). In this research, these composites' impact, flexural, and tensile characteristics were also studied. Both sets of findings were contrasted with those of a composite that was identical and had an ultraviolet ray

stabilizer and colorant mixture (UVC). Both sets of findings were contrasted with those of a composite that was identical and had a UVC. As a result of the research, the synergistic impact study demonstrates unequivocally that the addition of wool to the PP/KF/APP system improves the flame-retardant performance of the entire system by outperforming the performances of individual components during residual char generation.

2.2.3 Fabrication of wool in cement-based composites

Fantilli et al. (2017) investigated the use of wool as fiber reinforcement with the aim of using the wool in cement-based composites. The result showed the wool fibers that have been treated and those that have not are equivalent in terms of strength and elastic modulus to the most popular vegetal fibers used to enhance cementitious mortars. This clearly shows that the inclusion of wool fibers, whether or not they have been exposed to atmospheric plasma, can enhance cementitious mortar's functionality and extend the material's useful life. This behavior is comparable to that seen when cementitious mortars are reinforced with hemp fibers.

Ghaffar et al. (2020) investigated the mechanical interlocking of natural fiber-reinforced cementitious composites using 25 mm strands of uncoated and resin-coated wool twine introduced into different epoxy matrices. The primary goal of the study is to produce composites with balanced characteristics that have fibers dispersed uniformly throughout the matrix and strong interfacial bonds between the fibers and cement matrix. The creation of composites with well-balanced characteristics that have fibers dispersed uniformly throughout the matrix and strong interfacial bonds between the fibers and cement matrix. Apart from that, this research also aims to investigate different pozzolanic matrices with manipulated variable of mix ratios by weight. Additionally, flax/wool twine coated in epoxy and polyurethane resin is utilized to evaluate the physical and morphological characteristics of mortars in comparison to unreinforced (UNR) and uncoated samples. So, when compared to UNR samples, uncoated flax/wool twine may behave as a defect in the mortar, reducing the compressive and flexural strength. This is due to the poor dispersion of the flax/wool twine after it has been untangled into individual fibers during mixing. Furthermore, the flax/wool resin coatings improved the physical properties of mortar samples compared to nonreinforced and noncoated samples.

This study used an experimental design to assess the effects of various rock wool concentrations on the natural bond strength and bond stress-slip behavior of lightweight concrete-filled steel tubes (LCFTs) after thermal loading at high temperatures (Bahrami & Nematzadeh, 2021). The load-reversed push-out test involved up to four loading cycles for the LCFT specimens. The influence of rock wool on the LCFT specimens' heat-induced bond strength loss was investigated during the first loading cycle, and the macro locking component was assessed during the second, third, and fourth cycles after chemical adhesion and microlocking were eliminated. Finally in this research, a temperature-dependent bonding properties prediction model for the LCFT specimens, including rock wool waste, was developed.

2.2.4 Fabrication of mineral wool-based composites

In [Xie et al. \(2016\)](#), materials made of layer-structured mineral wool with broad-spectrum absorption characteristics have been created based on the impedance matching theory and EMW propagation laws. In a frequency range of 2–18 GHz, the reflection loss was examined, and the impact of the macro- and microstructure on the absorption qualities was studied. According to [Xie et al. \(2016\)](#), the mineral wool board developed in this study may be employed as architectural ornamental and fitting materials in building projects and can efficiently block indoor electromagnetic radiation. Granulated mineral wool was utilized as the substrate material in this experiment, and [Fig. 2.1](#) illustrates its chemical compositions.

As the result of the experiment in [Xie et al. \(2016\)](#), the findings of the electromagnetic parameters testing indicate that mineral wool is not very effective at attenuating electromagnetic waves, and carbon black (CB) is a type of material with a high dielectric loss.

In addition, another researcher studied the mineral wool composite with the intention of determining if Portland cement might be improved by adding the filter dust that is gathered during the manufacture of mineral wool ([Kubiliute et al., 2018](#)). The primary constituents of the dust under investigation in this study are quartz, albite, dolomite, and the amorphous phase. Most of the contaminants are halite and sylvite. Furthermore, the tested additive was milled and manufactured as a microfiller. The results from the research by [Kubiliute et al. \(2018\)](#) demonstrated that while extending the dormant time, the cupola dust addition boosts the first hydration of cement. After 28 days of hardening, it was calculated that the dust addition could replace up to 15 wt. % of Portland cement without negatively impacting the samples' strength qualities.

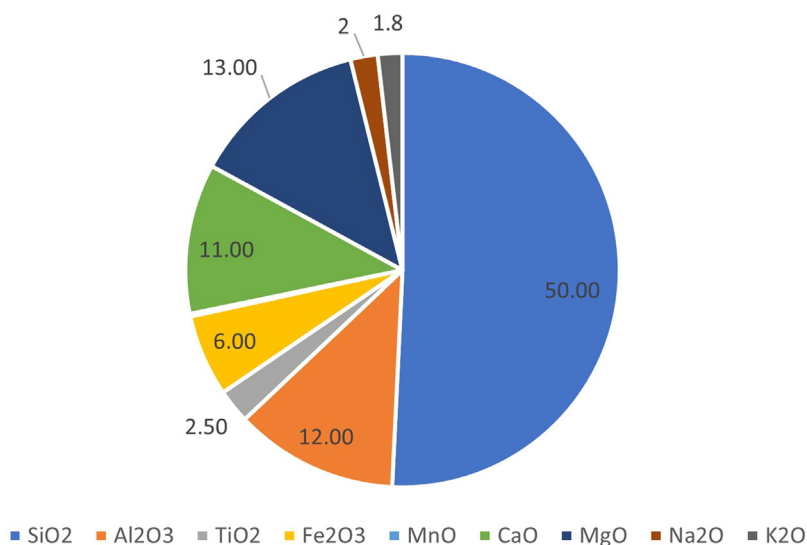


Figure 2.1 pie chart of different chemical composition present in mineral wool. The mineral wool, chemical composition.

Based on Piña Ramírez et al.'s (2019) research, to determine if these mortars can replace the less environmentally friendly traditional mortars, it is important to conduct a mechanical investigation of the mortars constructed from three distinct types of mineral wool waste. Apart from that, the highest quantity of fiber addition for recycling and achieving the necessary mechanical strengths comparable to those attained for cement matrix with commercial fibers are the two key goals. At the conclusion of the research, an effective replacement for the present reinforcing fibers in cement-based mortars is the use of mineral wool leftovers.

2.3 Characteristic of wool-based composites

The composition of a composite wool is wool fibers, synthetic polymer, and other natural fibers. The resulting material has properties that are different from those of individual components, and it is often used in a variety of applications, such as clothing, textiles, and construction materials (Santulli et al., 2013). Wool composites are known for their durability, strength, and resistance to wear and tear, making them a popular choice for a range of products. They can also be engineered to have specific properties, such as being fire resistant or water resistant, depending on the needs of the application.

2.3.1 Mechanical properties of wool-based composites

In terms of mechanical and other functional properties, the high stiffness and specific strength of certain fibers provide significant benefits (Pérez et al., 2012). Keratin is a special type of protein which is commonly found in feathers, nails, and wool. This protein is long-lasting, hydrophobic, inert chemically, and malleable. Keratin is distinguished from other protein molecules by its great physical and thermal resilience, which is obtained not just through hydrogen bonding and London forces but also by an increase in concentration of the amino acid cysteine (Keya et al., 2019). The complicated structure of the microfibril-matrix of wool, in fact, plays an important role in defining mechanical qualities (Tsobkallo et al., 2012).

2.3.1.1 Tensile strength

To identify the tensile strength of wool composite, tensile tests were carried out using an Instron 5567 UTM (10 kN load cell). Based on the ASTM D3882, the tensile moduli and strength were evaluated where the chord modulus is set to 0.25%–0.005% of strain. The crosshead speed is set to 12.5 mm/min. The video extensometer recorded the tension.

Flexural testing was performed on the Instron 5567 UTM using a 3-point bending rig based on ASTM D790 to determine the properties of a single wool fiber. The span length of 24 mm and crosshead speed of 16 mm/min were used as specified by the testing standard (Kim & Bhattacharyya, 2016). The outcome is shown in Table 2.1 based on the procedure.

Table 2.1 Properties of coarse wool fibers (Kim & Bhattacharyya, 2016; Das et al., 2017).

Properties	Diameter	Length	Density	Tensile modulus	Tensile strength
Recorded value	45.3 ± 0.11 mm	2.87 ± 1.15 mm	1.39 ± 0.11 g/cm ³	4.8 ± 0.7 GPa	160 ± 23 MPa
Evaluation method	Optical microscope/ image analysis		Archimedes method	Instron testing machine	

According to Das et al. (2017), the tensile properties of a single wool fiber were shown the similar properties, hence validating the results obtained by (Kim & Bhattacharyya, 2016).

Yükseloğlu (2015) conducted a similar research to identify the tensile strength of wool composite by using WFs in warp direction, as a ribbon form, together with WF waste, as a reinforced material. Tensile strength samples were prepared in accordance with TS 1398 EN ISO 527. The 4% WF with 2%, 4%, and 6% waste were evaluated. For each composition, the extension on maximum load and strain strength were measured. The results revealed that as the percentage of waste material grew, the maximum load extension decreased. Similarly, as waste content increased, so did strain strength. These findings shed light on the mechanical behavior of WF composites containing various percentages of waste material. Based on the result, it is shown that the wool can be used as the base of a composite. The tensile strength of the composites was decreasing linearly along wool waste. Tensile strength values derived from 4% WF + 4% waste fiber; mixed reinforced materials were the closest to the pure resin record. Hence, the amount of wool waste fiber drops within the reinforced composite constructions, elongation on the maximum load increases.

According to Porubská et al. (2015), following the analysis of gamma radiation up to 400 kGy, it was discovered that the tensile strength did not modify the initial extension, which first rose and then decreased monotonously. Natural fibers (wool) used as reinforcing properties in cement-based composites can improve concrete toughness. Similarly, mortar strength increases while delivering a more environmentally friendly alternative to typical industrial fibers. Indeed, such fibers can bridge the surfaces of fissures in the postfracture stages, lowering the construction industry's environmental impact (Fantilli et al., 2017).

2.3.1.2 Study of tensile test in sheep wool fibers

The impact of sheep wool fibers and fly ash on cement compressive was examined. Tensile test is also conducted to identify the yield strength of wool fiber (Grădinaru et al., 2022). Seven specimens were made, one of which was the reference concrete mix. In order to compare the properties of concrete with additions, 10% of the cement quantity was replaced with fly ash and sheep wool fiber in two lengths (20–30 mm and 50–60 mm), and two mass percentages (0.35% and 0.80% of the total mass of the concrete mix) were added. The results of the tests demonstrated that adding sheep

wool fibers to concrete did not enhance its hardness at the percentages of addition tested, whereas it resulted in a lower strength.

Compression strength was reduced by one-third when sheep wool fibers were used instead of reference concrete. The degree of reduction, however, varied depending on the length and concentration of the fiber. Compression strength was unaffected by a fiber length of 55 mm and a dose of 0.35%, whereas a higher concentration or a shorter fiber length reduced compression strength.

The physical strength of a cement mortar with varying lengths and quantities of wool fibers was investigated by another researcher. The specimen is set to have 13%, 23%, and 46% of wool fiber in comparison to the weight of cement (Fiore et al., 2020). They observed that, regardless of fiber number or length, wool-reinforced composite contributes greatly lower compressive load than the control wool cement composite. However, according to Fantilli et al. (2017), wool increases the flexural strength and ductility of cementitious mortars. Wool, like other natural fibers, enhances mortar's mechanical and ecological characteristics.

2.3.2 Thermal properties of wool-based composites

The properties of microfibril-matrix wool, which contain high concentrations of sulfur (3–4 wt.%) and nitrogen (15–16 wt.%) of amino acid groups, contributes to the good thermal properties of wool (Tsobkallo et al., 2012). Fig. 2.2 and Table 2.2 show the results of biochar and wool in cone calorimeter (Das et al., 2017).

Biochar is a type of charcoal that is produced by heating organic material, such as wood, in an oxygen-limited environment. Based on the result obtained, charcoal has good thermal resistance properties. However, despite lacking behind biochar for wool, wool can be an alternative source. When compared to biochar, the first quick burning of wool can be readily recognized as a sharp peak at around a quarter minute. Despite this, the dissociation of the microfibril matrix composition and the breakdown of the disulfide connections drove the formation of the char, and as the char of the wool accumulated, the HRR declined.

To identify the thermal properties of wool waste fabric composite, Yükseloğlu (2015) conducted an experiment on thermal conductivity with composite samples that were tested for 45 and 90 minutes on the THISYS THI01 thermal conductivity measuring tester. The result indicating the highest thermal conductivity for the sample 4% WF with 6% waste is 0.392 W/mK followed by 4% and 2% waste. Conduction heat transmission is affected by the heat conductivity of the material. It was discovered that as the waste proportion grows, so does the heat conductivity factor of the wool waste composites (Yükseloğlu, 2015).

2.3.3 Hydrophobic behavior

Hydrophobic behavior tests were performed in accordance with ASTM: D570 (Santulli et al., 2013). Three specimens were oven dried at 105°C for 1 day. At 23°C, the conditioned specimens were immersed in distilled water for 2 hours, 1 day, and 2 days. After the time limit had expired, the specimens were removed

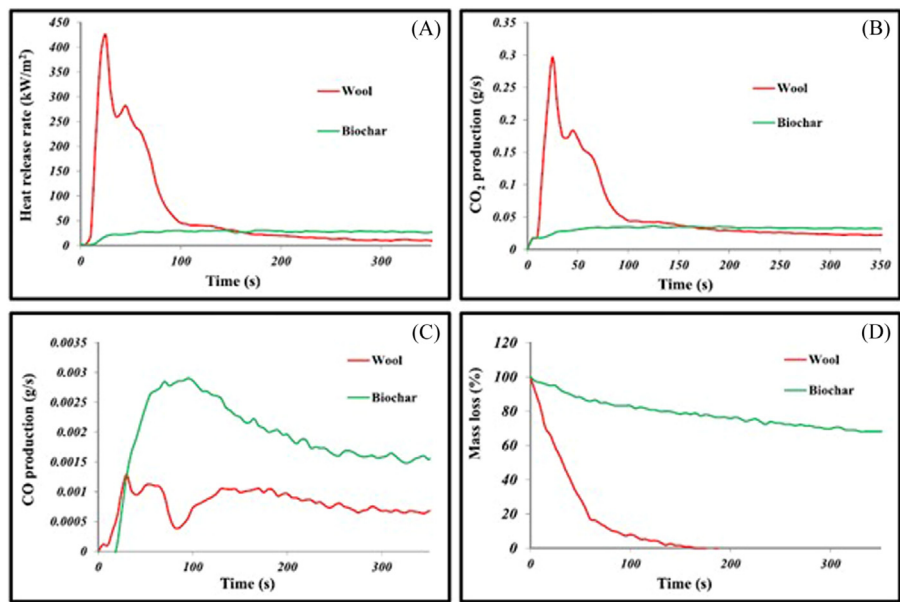


Figure 2.2 Results of biochar and wool in cone calorimeter: (A) heat release, (B) CO₂ production, (C) CO production, and (D) mass loss (%) with respect to time (Das et al., 2017). Source: From Das, O., Kim, N. K., Sarmah, A. K., & Bhattacharyya, D. (2017). Development of waste based biochar/wool hybrid biocomposites: Flammability characteristics and mechanical properties. *Journal of Cleaner Production*, 144, 79–89. <https://doi.org/10.1016/j.jclepro.2016.12.155>.

Table 2.2 Result obtained from the cone calorimeter (Das et al., 2017).

Specimen	Time of ignition/ TTI (s)	Time of PHRR/ TPHRR (s)	Peak heat release rate/ PHRR (kW/m ²)	Total of heat release rate/THR (MJ/m ²)
Wool	4.5 ± 0.7	25 ± 0.0	429.8 ± 6.5	19.6 ± 1.9
Biochar	—	305 ± 348	33.3 ± 4.6	9.2 ± 1.9

PHRR, Peak heat release rate; THR, total heat release rate; TPHRR, time to PHRR; TTI, time to ignition. Source: From Das, O., Kim, N. K., Sarmah, A. K., & Bhattacharyya, D. (2017). Development of waste based biochar/wool hybrid biocomposites: Flammability characteristics and mechanical properties. *Journal of Cleaner Production*, 144, 79–89. <https://doi.org/10.1016/j.jclepro.2016.12.155>.

from the water, all surface water was wiped away with a cloth, and they were weighed. The outcome is shown in Table 2.3.

The free hydroxyl groups in cellulose and hemicelluloses provide more sites for water absorption as natural fiber content increases. Water absorption in composites is influenced by the presence of lumens and hydrogen bonding sites in jute fibers and wool felts, gaps at the matrix-reinforcement interface, and porosity.

Table 2.3 Result of water absorption test (Santulli et al., 2013).

Specimen		W-33	W-40	J2W	J3W
Water absorption (%)	2 h	0.86 (0.03)	1.95 (0.09)	2.03 (0.62)	2.93 (0.47)
	1 day	1.77 (0.08)	4.19 (0.18)	4.93 (0.48)	6.70 (0.49)
	2 days	2.17 (0.11)	5.37 (0.21)	5.70 (0.44)	7.80 (0.57)

Source: From Santulli, C., Sarasini, F., Tirillò, J., Valente, T., Valente, M., Caruso, A. P., Infantino, M., Nisini, E., & Minak, G. (2013). Mechanical behaviour of jute cloth/wool felts hybrid laminates. *Materials and Design*, 50, 309–321. <https://doi.org/10.1016/j.matdes.2013.02.079>.

2.3.4 Toughness and morphology behavior

The endurance of wool-reinforced cement-based composites is affected by exploitation circumstances as well as external activities. The connection between the fibers and the cementitious matrix is critical (Fantilli & Józwiak-Niedźwiedzka, 2021). When combined with another material, such as a synthetic polymer, wool composites can be engineered to have specific properties and performance characteristics that make them well-suited for a variety of applications (Cardinale et al., 2017). For example, wool composites can be made to be highly durable and long-lasting, making them suitable for use in products that are expected to have a long lifespan, such as construction materials or industrial textiles.

Cement hydration was given as a critical element to comprehend the state of fiber deterioration, which is supported by test findings (Fantilli & Józwiak-Niedźwiedzka, 2021). The researchers studied the effects of the saline of Portland cement type 1 and the effects of curing period on the physical characteristics and morphology of reinforced mortars with sheep wool. The findings showed that when the saline element of the cement paste was lower, the wool fibers in the cement matrix had better resistance resulting in greater residual stress following fracture in wool-reinforced mortars.

Curing the mortar beams in distilled liquid at 27°C accelerated the degradation of wool fibers in the matrix formed with high-alkali cement concentration ($\text{Na}_2\text{Oeq} = 1.1\%$), in contrast to average- and low-alkali cement concentrations ($\text{Na}_2\text{Oeq} = 0.6\%$ and 0.4%). The morphology of specimens made with high-alkali cement is depicted in Fig. 2.3. Furthermore, the wool fibers were observed after 3 and 27 days in water at 20°C. The more damaged the wool fibers, the longer the time spent curing in water with high-alkali cement.

The salinity of Portland cement and the curing conditions were discovered to have a significant impact on the physical properties and morphology of wool-reinforced mortars as a result of the research. Lower salinity and room-temperature water curing can improve the resilience of wool fibers in the cement matrix, but higher saline properties and water curing can hasten wool fiber degradation.

2.3.5 Bio-based wool composites

Bio-based wool composites are a relatively new material that has been developed to merge the best of both natural and synthetic materials. This material is proving to

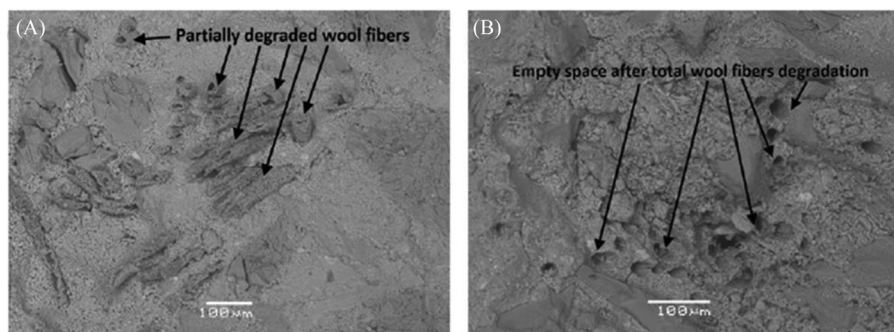


Figure 2.3 Scanning electron microscope (SEM) images of mortar. SEM observation of the mortar morphology: (A) 3 days, (B) 27 days, scale bar = 100 μm (Fantilli & Jóźwiak-Niedźwiedzka, 2021).

Source: From Fantilli, A. P., & Jóźwiak-Niedźwiedzka, D. (2021). Influence of Portland cement alkalinity on wool-reinforced mortar. *Proceedings of Institution of Civil Engineers: Construction Materials*, 174(3), 172–181. <https://doi.org/10.1680/jcoma.20.00003>.

be increasingly important for many applications due to its unique combination of properties. The use of bio-based wool composites for sustainable manufacturing has increased in recent years due to their numerous benefits. According to Zini and Scandola (2011), bio-based wool composites have low-density and high-strength properties that make them ideal for use in many applications. On top of this, they are also highly resistant to flame and have good thermal insulation, which can further reduce the energy consumption of the manufacturing process. Additionally, they are also highly recyclable, making them a great choice for sustainable manufacturing. Furthermore, they can be produced in a variety of shapes and sizes, making them extremely versatile and allowing them to be used in a wide range of applications. Finally, they are also biodegradable, meaning that they do not contribute to the accumulation of waste in the environment. All of these factors make bio-based wool composites an ideal choice for sustainable manufacturing applications (Zini & Scandola, 2011).

Bio-based wool composites have been found to provide unique properties to materials, allowing for more efficient and sustainable material production (Wool & Sun, 2011). As a natural fiber, wool is both lightweight and strong, making it an ideal component for composite materials. The wool fibers add flexibility and strength to the material, allowing for more efficient production processes and a greater range of applications. Additionally, wool is also an environmentally friendly material, as it is biodegradable and has a carbon footprint that is significantly lower than that of synthetic materials. This means that bio-based wool composites can be used to produce materials that are more sustainable and efficient. In addition to this, wool composites also provide increased insulation and energy efficiency, which can help reduce energy costs and improve the overall performance of materials. Overall, the impact of bio-based wool composites on material properties is substantial, making them an ideal component for a variety of applications (Wool & Sun, 2011).

Bio-based wool composites have excellent physical and mechanical properties that make them an attractive option for use in a wide range of applications. The composites have superior tensile strength, sound and acoustic dampening, electrical insulation, long-term thermal stability, and excellent dimensional stability (Hong & Wool, 2005). Furthermore, the composites are nontoxic, fire retardant, and recyclable, which make them a desirable option for green materials. The combination of these properties makes bio-based wool composites a promising material for a variety of applications, ranging from automotive to industrial.

2.4 Application of wool-based composites

2.4.1 Cement-based composites from sheep wool

The most common construction material, concrete, has low tensile and flexural strengths. Steel, glass, basalt, polyester, hemp, PP, nylon, bamboo, human hair, and animal wool fibers can be used to improve impact strength, specific stiffness, ductility, tensile strength, shrinkage, and cracking resistance in cementitious matrix. Most of those are artificial, emit carbon, and are expensive. Natural fibers from various sources are recycled and used as environmentally friendly building materials, attracting research attention due to their lightweight, low-cost, nontoxic, easy to process, recyclable, and renewable properties (Alyousef et al., 2020). In the same way, animal fibers like sheep's wool are being marketed as building insulation. Sheep produce between 2 and 4 kg of raw wool per year, which must be sheared for their health. Textile industry cannot use 75% of European sheep farms' wool. Before disposal, it must be sterilized at 130°C as special waste. This unused wool, plus the wool in other end-of-life products and textile industry waste, totals 150 million tons per year (Fantilli et al., 2017).

Sheep wool is renewable, eco-friendly, and recyclable, making it a popular building material. Wool is nonflammable. They are safer to burn than air because they need more oxygen. They burn without melting, dripping, or sticking to the skin. It carbonates and does not burn. Its nitrogen content prevents combustion. Wool-reinforced mortars had higher postcracking residual stresses because wool fibers were more resistant to cement paste with lower alkalinity. Wool reinforcement performs similarly to PP. However, improving the workability of sheep wool fiber-reinforced concrete can greatly improve its fresh and hardened properties. Due to their mortar-like composition, wool-reinforced composites, as shown in Fig. 2.4, are ideal for renovating historic buildings (Jóźwiak-Niedźwiedzka & Fantilli, 2020).

2.4.2 Waste rock wool as oil absorbent

Oil contaminants in water are complex mixtures of aliphatic (C_nH_{2n+2}), naphthenic (cycloalkanes), and aromatic hydrocarbons. There are many oil reserves in water, the most significant of which are oil extraction and transportation facility leaks, natural gas seeps from the seafloor and seabed, seafloor drilling, emergency floods (tank collisions

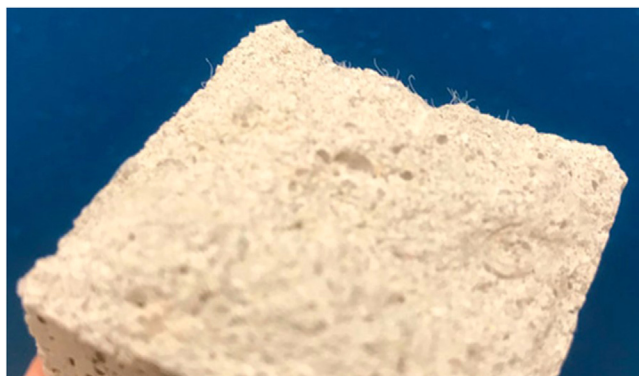


Figure 2.4 Mortar image containing wool fibers. A sample of the modified mortar containing wool fibers (Pederneiras et al., 2019).

Source: From Pederneiras, C. M., Veiga, R., & de Brito, J. (2019). Rendering mortars reinforced with natural sheep's wool fibers. *Materials*, 12(22), 3648. <https://doi.org/10.3390/ma12223648>.

or breakdowns), inland and marine navigation, air and ground transportation, waste petroleum-based industrial installations, and urban stormwater (Gohub & Piekutin, 2018). Effective oil or water separation from oily wastewater requires membrane material with special wettability. Superhydrophilic hollow fiber membranes with superior antifouling properties efficiently separate oil and water (Wang et al., 2019).

Rock wool, a basalt or dolomite fabric, is inexpensive and noncombustible. It is used in fireproofing, sound absorption, and cement reinforcement. The majority of rock wool waste was dumped or buried, causing environmental issues. New waste rock wool application techniques are thus required such as waste rock wool nanocomposite for the removal of Cr(VI) from liquid and soil. Yap et al. (2021) study enhanced the recycling of rock wool (Fig. 2.5). Rock wool is an effective water separator to oil due to its low cost and high porosity. Rock wool is also heat resistant. Heating reduces the viscosity of the oil, making separation easier. Superhydrophobic rock wool heated with an electric current separates high-viscosity oil from liquid. Pure rock wool is ineffective at separating oil and liquid due to its hydroxyl groups. The hydrophobicity must be altered. A simple dip-coating technique was used with environmentally friendly PDMS/SiO₂ modifying agents. Rock wool is extremely hydrophobic where it successfully separates oil and liquid. When heated with an electric current, modified rock wool separated high-viscosity oil continuously due to its heat resistance. This low-cost, environmentally friendly oil absorbent can be used to treat high-viscosity oil spill pollution (Hao et al., 2019).

2.4.3 Waste rock wool as hydrogen sulfide gas absorbent

Laboratory research indicates sulfide concentrations of up to 63,000 ppm in landfill gas from plasterboard demolition waste. Hydrogen sulfide, also known as sulfide or

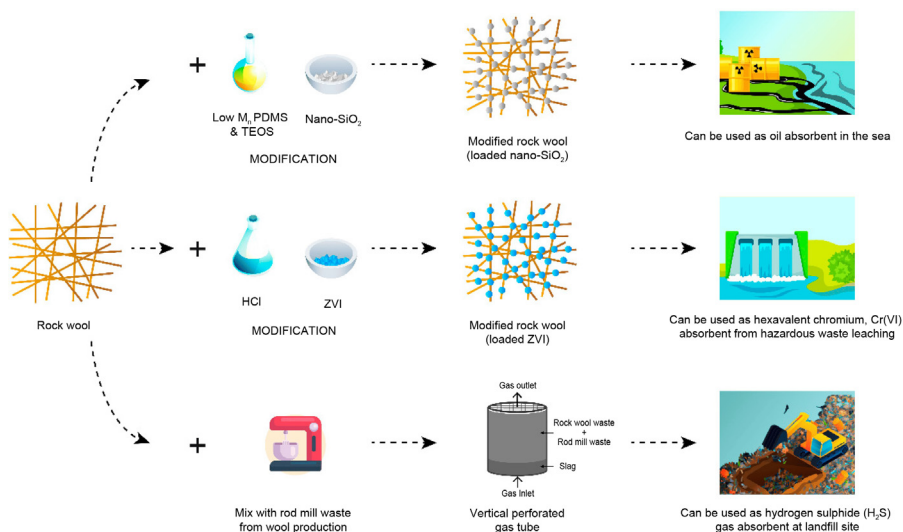


Figure 2.5 Waste rock wool and its modification and applications. The use of waste rock wool (Yap et al., 2021).

Source: From Yap, Z. S., Khalid, N. H. A., Haron, Z., Mohamed, A., Tahir, M. M., Hasyim, S., & Saggaff, A. (2021). Waste mineral wool and its opportunities—A review. *Materials*, 14(19), 5777. <https://doi.org/10.3390/ma14195777>.

H₂S, with an odor threshold of 0.1–0.5 parts per billion, is an important landfill odorant. Sulfur-containing wastes continue to accumulate despite a decade of waste management regulations, including an organic waste ban and increased reprocessing of many ends up wasting fractions such as plasterboard and other sulfur-containing demolition waste material. An experience from Norwegian municipal solid waste (MSW) landfills shows that concentrations above 5000 ppm H₂S are common, with extreme levels reaching 30,000 ppm (Bergersen & Haarstad, 2014).

Initially, rock wool was used to absorb sulfur gas and polycyclic aromatic hydrocarbons. Several studies have shown that waste rock wool can absorb toxic and corrosive biogas, as well as hydrogen sulfide (H₂S). Biogas is used to generate energy because it contains methane, carbon dioxide, and a trace of hydrogen sulfide. Combustion engines are harmed by H₂S gas. As a result, biogas contaminants, particularly hydrogen sulfide, must be removed. Waste rock wool was mixed with rod mill waste in a 19-to-1 ratio to create a vertical perforated gas tube. The rock wool filter removed 88% and 95% of the sulfide gas administered in a downward flow over the course of 80 days (Yap et al., 2021).

2.4.4 Biomaterials from wool keratins as regenerative medicine

“Keratin-based materials” shaped the modern biomaterials due to their mechanical durability, biodegradability, and biocompatibility. For biomedical applications, they can

be cast as films, sponges, and hydrogels. Hair, wool, feathers, horns, and nails are mostly keratin. Structure and properties vary widely in these biopolymers. The most underexploited protein sources such as wool, hair, horns, feathers, and nails are used to extract keratin. Multicell wool fibers are sulfur rich. The top five wool producers are Australia, China, New Zealand, Iran, and Argentina. Two to three million tons of wool and 65 million tons of feathers are produced annually (Feroz et al., 2020).

Pharmaceutical, medicinal, cosmetic, and biotechnology product development heavily involves keratin. Keratin from wool can be made into mats, sponges, porous foams, films, sheets, gels, microfibers, agricultural films, bulk materials, and drug coatings as shown in Fig. 2.6 (Ranjit et al., 2022). These bio-based, adaptable materials are being studied. Nonantigenic keratin is useful in wound healing, tissue restoration, diffusion and cell seeding, and drug delivery as a topical or implantable biomaterial. Keratin biomaterial can be implanted as a film, sheet, or scaffold to provide structural integrity and stability under mechanical load, then break down to leave neo-tissue (Allafi et al., 2022). Different wool dissolving methods and different applications of wool dissolving products are summarized in Fig. 2.7.

2.4.5 Sheep wool as fertilizer

Sheep wool is a fibrous protein substance. Keratin is biodegradable and hygroscopic where it absorbs heavy metals. Waste wool contains Fe, Na, P, and K, which chelate

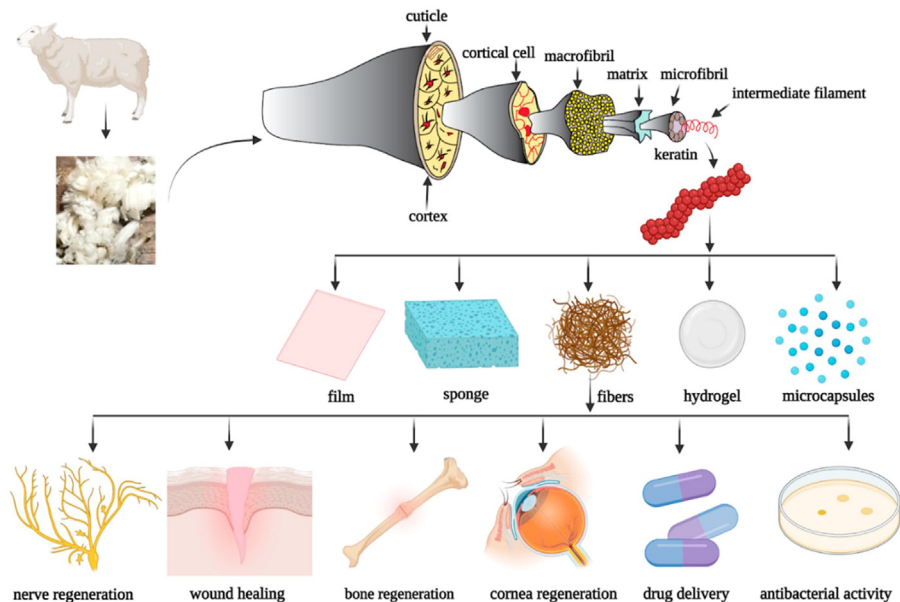


Figure 2.6 Tissue engineering using keratin-based wool materials (Ranjit et al., 2022).

Source: From Ranjit, E., Hamlet, S., George, R., Sharma, A., & Love, R. M. (2022).

Biofunctional approaches of wool-based keratin for tissue engineering. *Journal of Science:*

Advanced Materials and Devices, 7(1), 100398. <https://doi.org/10.1016/j.jsamd.2021.10.001>.

with plant roots and help them grow. Wool hydrolysate complex amino acids may regulate plant development. Wool is used in fertilizers, microcapsules, medicinal devices, and composites due to these qualities. Hydrolysis uses enzymes, acids (H_2SO_4 , HCl , and HCOOH), bases (CaO , NaOH , and KOH), a liquor at high temperature and pressure, and superheated water pumped into the hydrolysis reactor to achieve the appropriate physical conditions (Bhavsar et al., 2016). Some comprehensive method offered for valorizing agricultural and wool wastes is shown in Fig. 2.8.

Nitrogen fertilization may decrease vegetable productivity. The restricted availability of approved and inexpensive fertilizer materials in organic management systems raises production costs and reduces crop yields and farm profitability (Bradshaw & Hagen, 2022). Sheep wool, especially protein hydrolysate, has been recommended

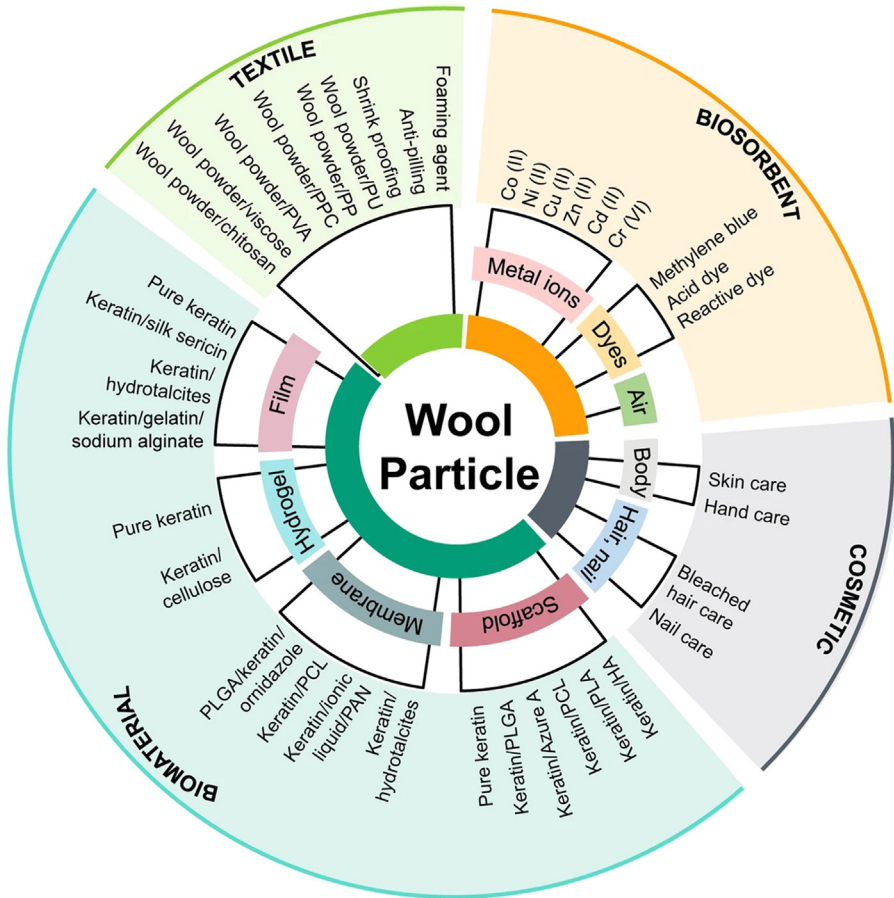


Figure 2.7 Applications of wool particle (Zhang et al., 2020).

Source: From Zhang, C., Xia, L., Zhang, J., Liu, X., & Xu, W. (2020). Utilization of waste wool fibers for fabrication of wool powders and keratin: A review. *Journal of Leather Science and Engineering*, 2(1), 15. <https://doi.org/10.1186/s42825-020-00030-3>.

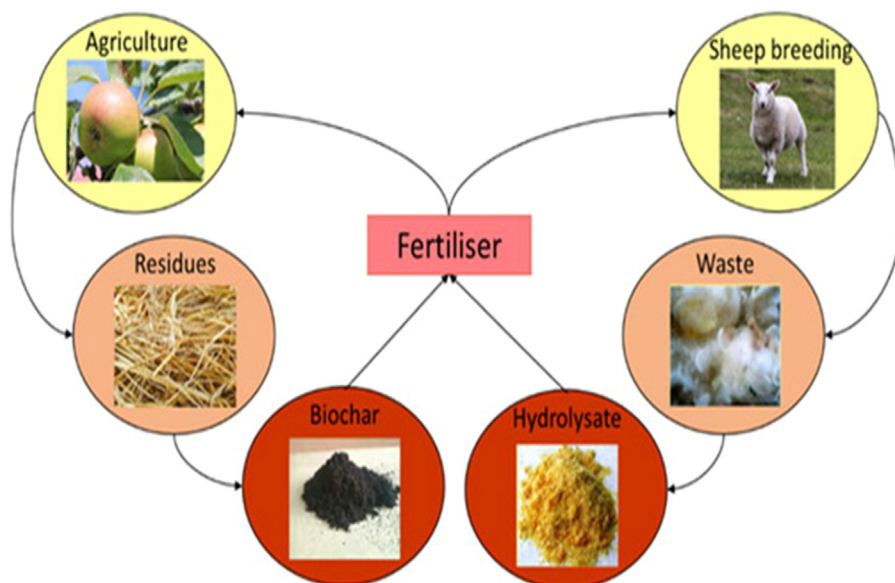


Figure 2.8 Different method of agricultural and wool wastes. The comprehensive method offered for valorizing agricultural and wool wastes (Marchelli et al., 2021).

Source: From Marchelli, F., Rovero, G., Curti, M., Arato, E., Bosio, B., & Moliner, C. (2021). An integrated approach to convert lignocellulosic and wool residues into balanced fertilisers. *Energies*, 14(2), 497. <https://doi.org/10.3390/en14020497>.

as a fertilizer ingredient. Sheep wool waste is a nitrogen-rich fertilizer for crop plants. By boosting soil N, C, and P, sheep wool hydrolysate improves plant growth. Hydrolyzed wool helps plant growth by chelating soil microelements. Overall, waste wool shows potential as a fertilizer, but a sustainable approach is needed to overcome all the issues and consider industrial-scale utilization (Allafi et al., 2022).

2.5 Conclusions

This chapter has discussed the synthesis, characteristics, and application of wool-based composite. The wool-based composite is useful for the environment in such that the manufacturing of concrete composites utilizing recycled household and industrial wastes and preservation of a cleaner environment were both addressed using sheep wool fibers. This clearly shows that the sheep wool fibers play a vital role in having a cleaner environment. In addition, the World Business Council for Sustainable Development estimates that the cement sector contributes 5%–7% of the world's total man-made CO₂ emissions. In addition, the World Business Council of Sustainable Development estimates that the cement sector contributes 5%–7% of the world's total man-made CO₂ emissions. Therefore sheep's wool is one animal fiber that has started

to be sold and promoted as a substitute for traditional insulating materials in building construction. Wool-based composite is also categorized as sustainable energy since its raw ingredients are readily recyclable, renewable, accessible locally, and ecologically benign. Wool-based composite has an important role on solar thermal system due to the utilization of sustainable energy in the system. For example, to increase the system's thermal efficiency, aluminum wool has been employed. This clearly demonstrates that wool-based composites, which are also classified as sustainable energy, play an important role in all developments.

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